B224-101-1112

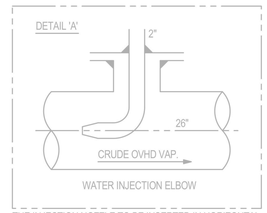
101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C 4
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

FLARE 101-1177
4" -FL-101-1204-A11A
3"-FL-101-1205-A11A
4"x3"
3"-WP-101-4811-B19A-IH
FROM 101-E-130A/B WASH WATER 101-1148
FROM 101-C-101 CR COL OVHD VAP 101-1127
48"-P-101-2701-A19A-IH
NOTE-3,5
NO POCKETS
MAKE CONNECTION AT MID POINT OF HEADER FROM TOP
TW 1212
PI 1205
TE 1223
TT 1223
TI 1223
FF
MIN.
36"-P-101-1210-A19A-IH
NOTE-3,5
Line 7
Line 8
36"x40"
48"x40"
40"x36"
48"x40"
NOTE-3,5
36"-P-101-1211-A19A-IH
Line 6
TI 1205
TT 1205
PI 1201
TW 1210
OP
Line 9
Line 10
24"-P-101-1209-B1A-IT TO 101-E-103A/B & 105A/B
Line 11
101-1113
14"x20"
20"x24"
Line 13
NOTE-6
Line 14
Line 15
Line 12
NC
14"-P-101-1209-B1A-IT
FI 1201 A
FI 1201 B
FT 1201
FE-1201
26"-P-101-1212-A19A-IH
FACING
SLOPE 1:500
TI 1201
TT 1201
TE 1201
12"-P-101-1205-B1A-IT
2"-SV-101-1201-B1A-IT
1.5E2 36.7
PSV 1201
900# 2"x1-1/2"
3/4"
LO FB
2"-WP-101-1201-B19A-IH
4 SP 036
NOTE-4
B19A A19A
A11A A19A IH
2"-P-101-1220-B1A-IT
MIN.
26"x24"
T1 T2 S1 S2
300# 300#
A
FI 1202 A
FI 1202 B
FT 1202
FE-1202
TI 1202
TT 1202
TE 1202
12"-P-101-1206-B1A-IT
2"-SV-101-1202-B1A-IT
2"-WP-101-1202-B19A-IH
3"-FL-101-1202-A11A
1.5E2 36.7
PSV 1202
900# 2"x1-1/2"
3/4"
LO FB
2"-P-101-1221-B1A-IT
B
SP 110
NOTE-4
A19A B19A
CP 1210
SP 010
CC
2"
3/4"
A19A A11A IH
26"-P-101-1213-A19A-IH
SLOPE 1:500
2"-WP-101-1202-B19A-IH
FI 1203 A
FI 1203 B
FT 1203
FE-1203
TI 1203
TT 1203
TE 1203
12"-P-101-1207-B1A-IT
2"-SV-101-1203-B1A-IT
1.5E2 36.7
PSV 1203
900# 2"x1-1/2"
2"-WP-101-1203-B19A-IH
SP 111
NOTE-4
2"-P-101-1222-B1A-IT
C
FI 1204 A
FI 1204 B
FT 1204
FE-1204
TI 1204
TT 1204
TE 1204
12"-P-101-1208-B1A-IT
2"-SV-101-1204-B1A-IT
1.5E2 36.7
PSV 1204
900# 2"x1-1/2"
26"-P-101-1215-A19A-IH
SLOPE 1:500
2"-WP-101-1204-B19A-IH
3"-FL-101-1204-A11A
SP 112
NOTE-4
NOTE-8
2"-P-101-1223-B1A-IT
D
SP 016
CP 1212
SP 018
CP 1208
101-E-102A/B/C/D
TT 1206
TE 1206
TI 1206
PI 1209
NOTE-8 (TYP.)
SP 008
CC
CP 1205
12"-P-101-1201-B1A-IT
12"-P-101-1202-B1A-IT
TE 1207
TT 1207
TI 1207
PI 1206
SP 012
CP 1206
TI 1208
TT 1208
TE 1208
PI 1212
SP 014
CP 1207
12"-P-101-1203-B1A-IT
12"-P-101-1204-B1A-IT
TE 1209
TT 1209
TI 1209
PI 1215
SP 018
CP 1208
FROM 101-E-101A/B CRUDE 101-1111
24"-P-101-1113-B1A-IT
NOTE-6
16"x12"
24"x16"
TWO PHASE FLOW
NOTE-5,10
NO POCKETS
24"-P-101-1219-A19A-IS
CR COL OVHD 101-1128
TO 101-A-101 D
24"-P-101-1218-A19A-IS
TO 101-A-101 C
24"-P-101-1217-A19A-IS
TO 101-A-101 B
24"-P-101-1216-A19A-IS
TO 101-A-101 A

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
4	10.03.22	ISSUED FOR CONSTRUCTION	SC	MW	YD	PV
3	30.09.21	ISSUED FOR CONSTRUCTION	SC	MW	YD	PV
2	02.09.20	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	RMB	KG	RS	SG
1	11.08.20	ISSUED FOR ENGINEERING/DESIGN	JO	KG	RS	SG
0	13.03.20	ISSUED FOR REVIEW	LC	KG	RS	SG

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
1	14.12.18	REVISED AS MARKED	PP	UA/AT	GN	AM
0	15.11.18	ISSUED FOR PACKAGE	S.K.L	UA/AT	GN	AM

PMC : इंजिनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) / ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

OWNER: एच पी सी एल राजस्थान रिफाइनरी लिमिटेड / HPCL RAJASTHAN REFINERY LTD / RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमेन्टेशन डायाग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

अनुमाप SCALE	कार्य संख्या JOB NO.	विभाग DEPT.	अनुभाग SECTN.	इकाई UNIT	आरेख संख्या DWG. No.	संशो REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. B224-CDU-TPL-101-1112 — ची पे एल आरेख संख्या — संशो REV 4

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DOCUMENT REVIEW DETAILS
1 No Comments. Doc. B224-CDU-TPL-101-1112 Rev. 4
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Returned for Records. V Void.
Signature with Date 24-AUG-2022
Name Reviewer UMER AFZAL (C889)
Page No. 1 of 2

Page 1 of 2